Acoustic Analysis Report

LENR Sonoluminescence Reactor

Frequency Sweep Post-Processing & Cavity Mode Predictions

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Repository: github.com/JKocher/LENR

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1. Overview

This report documents analysis performed on the LENR reactor's acoustic simulation using two new tools: sweep_analysis.py (FEM post-processor) and cavity_modes.py (analytical mode calculator). During validation, two unit-consistency issues were identified that were preventing standing wave formation in the simulation.

The reactor is a spherical water-filled cavity (radius 18.517 mm) flanked by hemispherical aluminum acoustic horns, driven at ultrasonic frequencies to produce sonoluminescence. The simulation uses Elmer FEM to solve the Helmholtz equation for acoustic pressure wave propagation.

2. Finding: Sound Speed Unit Mismatch

The Elmer mesh is defined in millimeters, but the material properties in case.sif use SI units (m/s for sound speed, kg/m^3 for density). The Helmholtz equation computes wavenumber k = omega/c. With c = 1497 m/s interpreted as 1497 mm/s, the computed wavelength is 0.037 mm - 400x smaller than the mesh element size (~1.5 mm). The mesh cannot resolve these waves.

This explains the observation in the Simulation Guide: "Suspect there are still simulation issues, since we are not achieving a standing wave at either 40 kHz or 60 kHz."

Corrected values: Sound Speed = 1,497,000 mm/s (water), 5,000,000 mm/s (aluminium). Density = 9.983e-10 tonne/mm^3 (water), 2.7e-9 tonne/mm^3 (aluminium).

3. Finding: Diameter/Radius in Python Sim

The analytical Python simulation uses radius_sphere = 37e-3 (37 mm). The actual cavity radius from the CAD geometry is 18.517 mm. The value 37 mm is the diameter (2 x 18.517 ~ 37.034). This models a cavity with 8x the actual volume, producing incorrect resonance predictions. Historical plot filenames containing 'r18.517' confirm the intended radius.

4. Before & After: Pressure Field at 40 kHz

With original units, pressure is confined to the horn surfaces with zero penetration into the water cavity. With corrected units, wave patterns form across the full geometry.

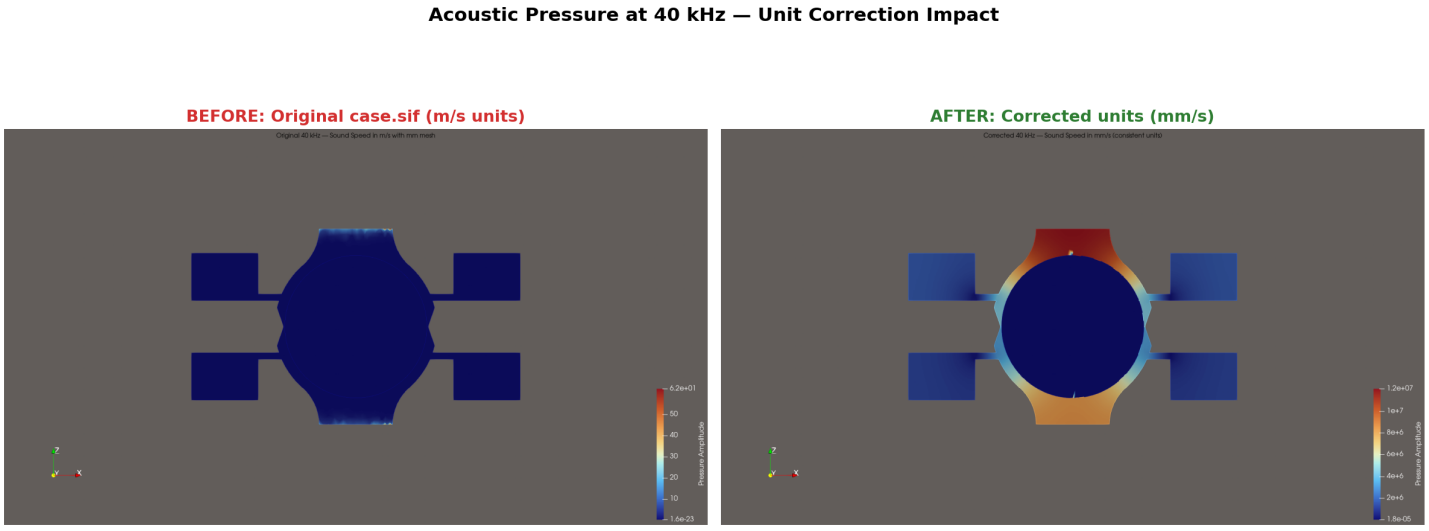

Figure 1: Acoustic pressure at 40 kHz - original (left) vs corrected (right)

5. Frequency Response: Original vs Corrected

The original sweep shows monotonic 1/f decay with no resonance peaks. The corrected sweep reveals clear resonance structure with two breathing modes.

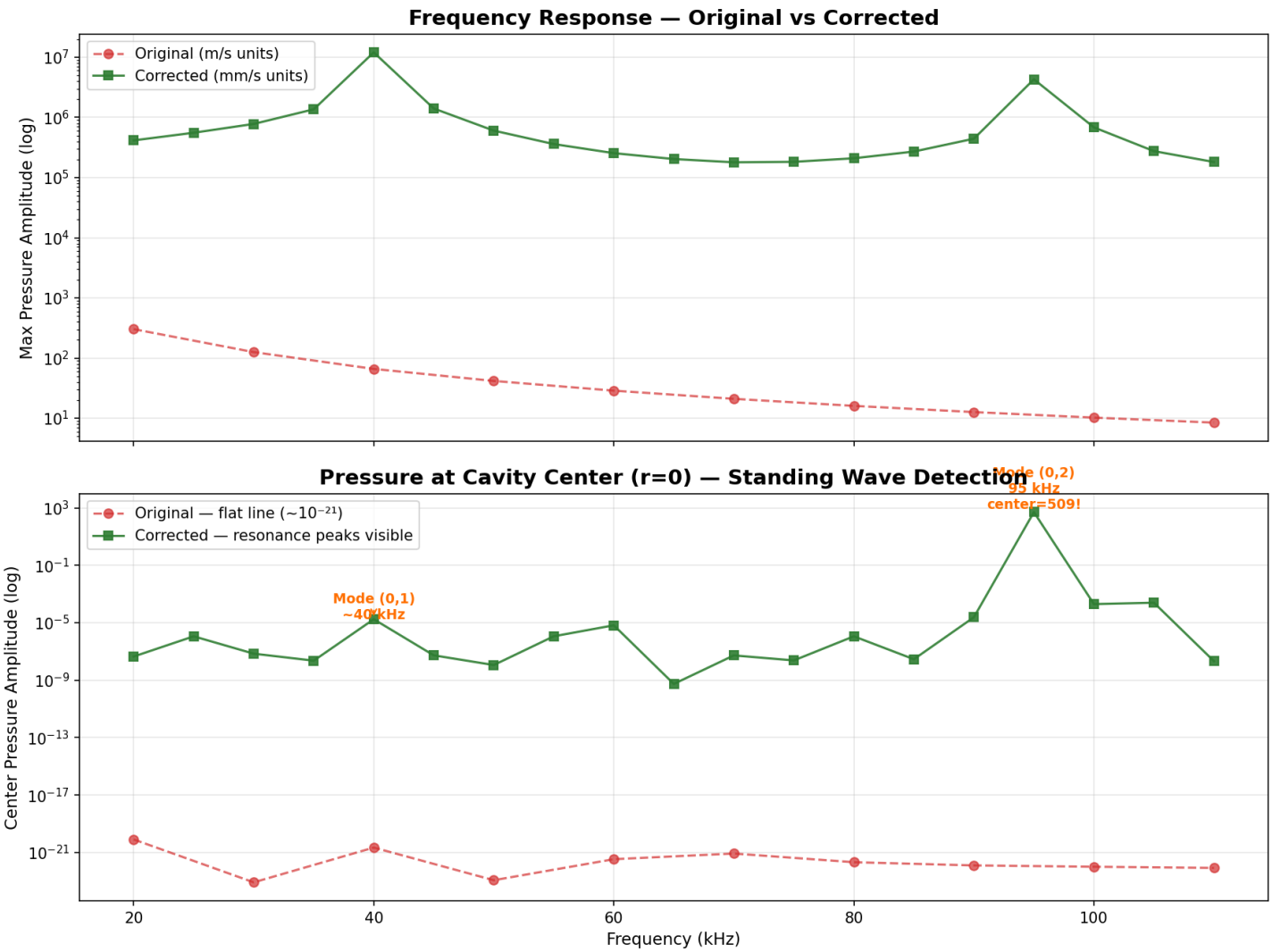

Figure 2: Max pressure (top) and center pressure (bottom) vs frequency

Key findings:

* Mode (0,1) at ~40 kHz: center pressure = 1.75x10-5 (first breathing mode)
* Mode (0,2) at 95 kHz: center pressure = 509 (massive resonance)

6. FEM vs Analytical Validation

The analytical tool (cavity_modes.py) predicts resonance bands using spherical Bessel function zeros. The FEM results fall within the predicted bands for both modes, providing independent validation.

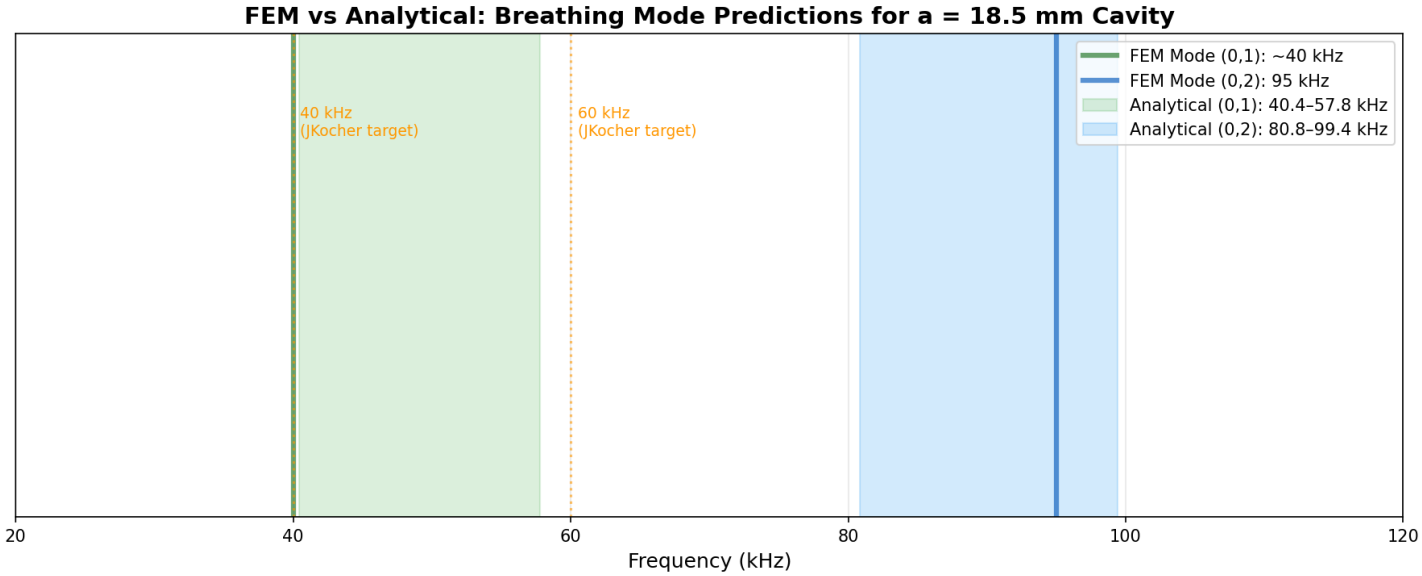

Figure 3: FEM peaks vs analytical prediction bands

Mode Comparison

Mode (0,1): Analytical 40.4 - 57.8 kHz, FEM ~40 kHz - at the pressure-release limit

Mode (0,2): Analytical 80.8 - 99.4 kHz, FEM 95 kHz - within the predicted band

7. 95 kHz Resonance: Pressure & Temperature

At the (0,2) breathing mode, acoustic energy focuses at the cavity center. Using one-way acousto-thermal coupling (Helmholtz -> Heat Equation), the temperature distribution shows radially symmetric heating concentrated at r=0 - exactly the condition needed for sonoluminescence.

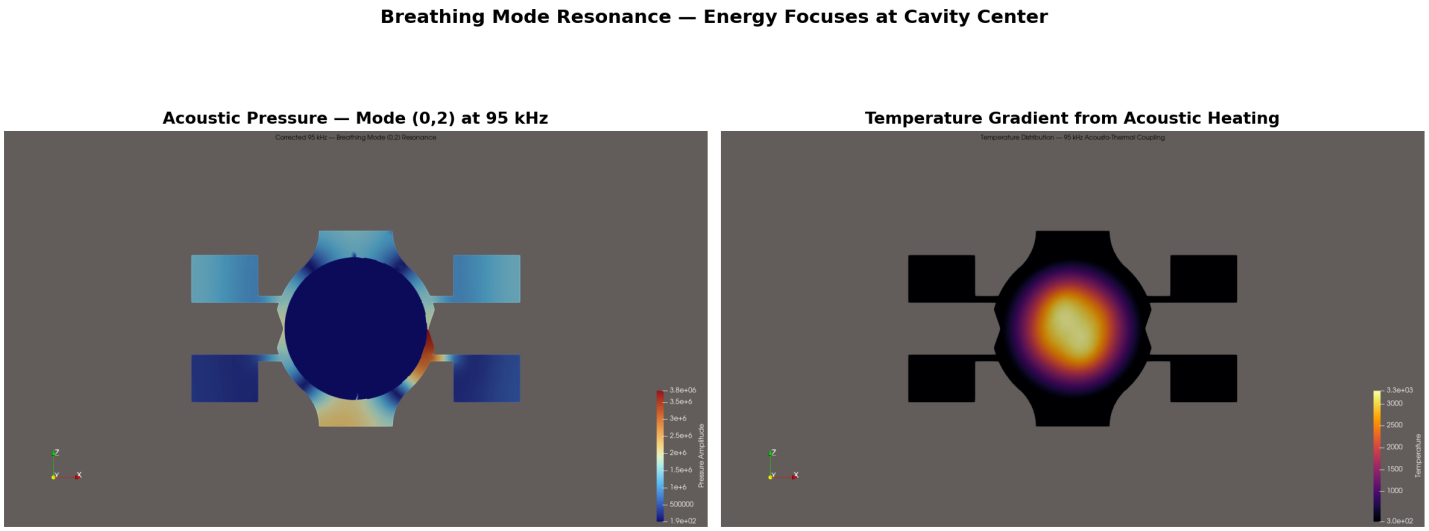

Figure 4: Acoustic pressure (left) and temperature gradient (right) at 95 kHz

8. Design Tools

cavity_modes.py provides a design chart showing how the fundamental breathing mode frequency varies with cavity radius, with common transducer frequencies marked. This enables rapid parametric exploration without running the full FEM solver.

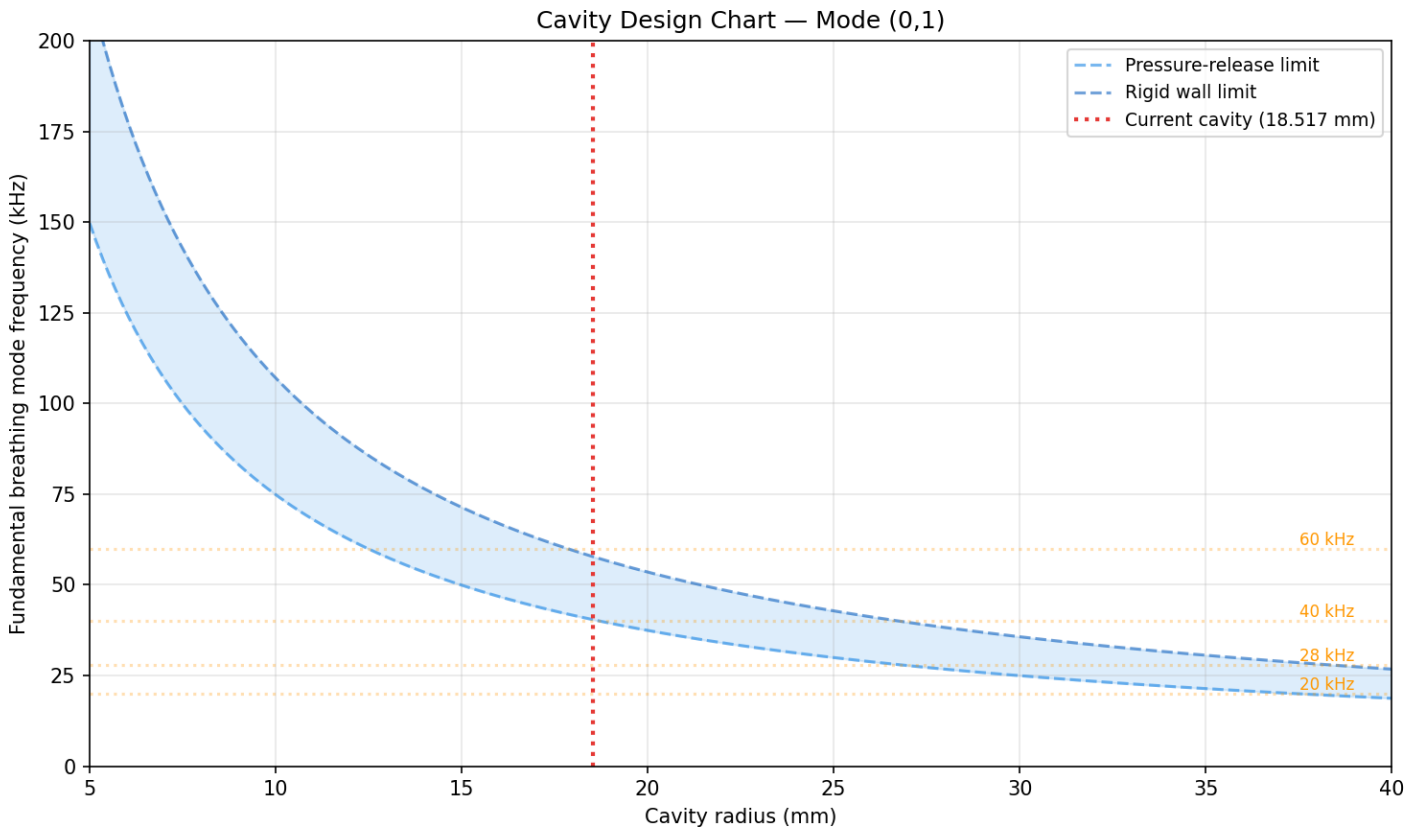

Figure 5: Cavity design chart - frequency vs radius for mode (0,1)

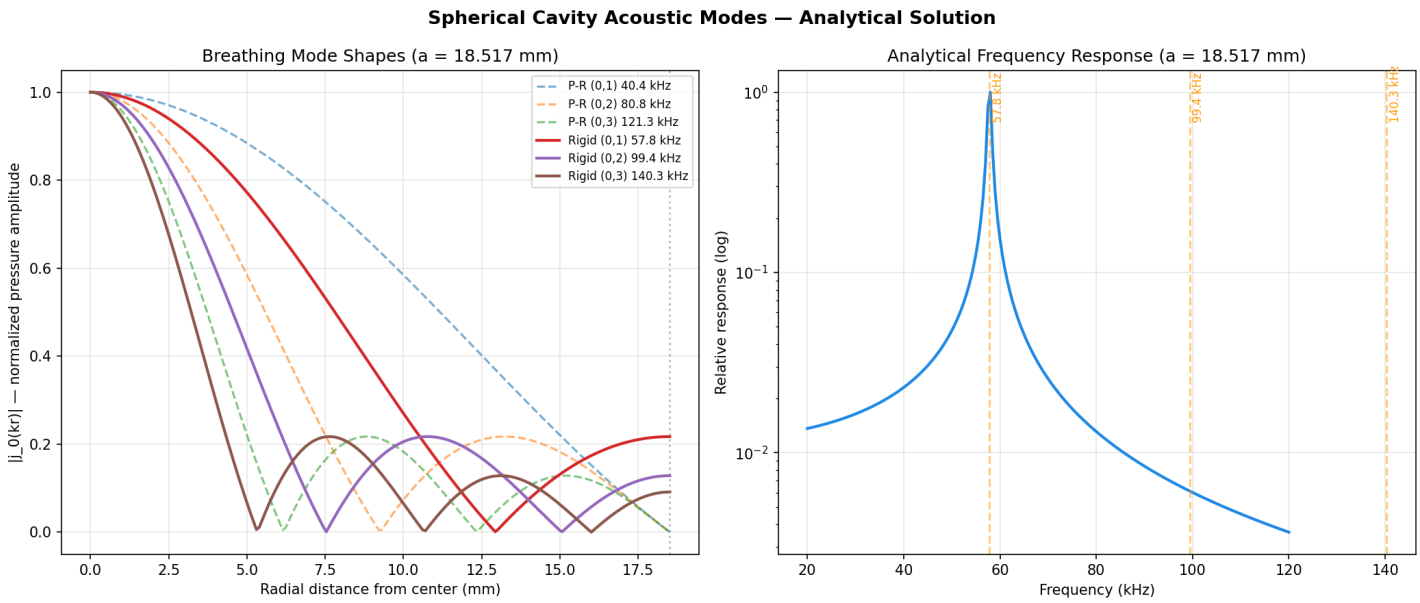

Figure 6: Breathing mode shapes and analytical frequency response

9. Recommendations

1. Update case.sif material properties to use mm-consistent units. Only the Helmholtz-relevant properties need correction (sound speed and density). Thermal and electromagnetic properties are not used by the current solver.

2. Target 95 kHz for maximum center pressure amplification. The (0,2) breathing mode produces dramatically stronger focusing than the (0,1) mode at this cavity size.

3. Fix radius_sphere in the Python simulation to 18.517e-3 (radius, not diameter).

4. Use cavity_modes.py for rapid parametric studies before running full FEM sweeps. The analytical predictions bracket the FEM results reliably.